Machine learning for sports betting

NBA and English Premier League: accuracy, calibration and archived-price returns.

Aman Dhillon • Panagiotis Housos

Supervisor: Daniel Totouom-Tangho

Statistical Arbitrage | Final project | 23 October 2026

PART 1

Methodology

The shared prediction-to-betting framework

PART 2

NBA results

Aman Dhillon

PART 3

Premier League

Panagiotis Housos

Sources: ML_Sports_Betting_Project_Proposal.pdf | Full paths in notes

Common methodology: from past games to bets

The proposal compares calibration-based and accuracy-based model selection. Implementation differs where the sports require it.

1 Earlier games

Earlier team stats
Prior-season strength
Player/xG extensions

2 Probabilities

NBA: home / away
EPL: home / draw / away
Exclude target inputs

3 Model selection

Accuracy branch
Calibration-error branch
Later-season evaluation

4 Betting rule

Compare probabilities with archived odds
Apply explicit stake rules

Estimated return per unit staked = model probability × decimal odds − 1

We test whether better probability estimates improve betting outcomes.
A higher classification accuracy or smaller calibration error does not guarantee a profitable price.

Sources: Data.ipynb; METHODOLOGY.md; ML_Sports_Betting_Project_Proposal.pdf | Full paths in notes

Common metrics; explicit league differences

Compare outcomes within each league. Keep the financial denominators and funding conventions separate.

Setting	NBA implementation	Premier League implementation
Outcome space	Home / away	Home / draw / away
Calibration criterion	20 bins; score set to 1 if fewer than 16 bins are occupied	Mean classwise ECE over 3 classes; 10 bins; no occupancy penalty
Post-fit calibration	No separate held-out temperature fit	Both initial branches receive held-out temperature scaling
Stake and starting cash	100 fixed or 1/8 Kelly; start 10,000	1,000 fixed or capped Kelly; start 100,000 in expanded accounts
Funding	Game-by-game settlement	Joint date-start cash allocation before settlement

Capital return = profit / initial cash. **Stake ROI (yield) = profit / total money staked.**

The NBA source calls capital return “ROI”. We relabel it here. Forecast accuracy is also different from the hit ratio of selected bets.

Sources: Data.ipynb; metrics.py; METHODOLOGY.md | Full paths in notes

NBA: data and model selection

Team box-score differences are lagged; each team's first ten appearances are excluded as prediction targets.

ELIGIBLE GAME ROWS

14,471

Across the supplied historical sample

FEATURE SCREENING

26 → 25 → 9

Candidate → drift-screened → nonredundant

BETTING EVALUATION

1,074

2018/19 games with odds

Selection branch	Chosen model	Selected inputs	2017/18 selection criterion
Calibration	Logistic regression	2	Penalised ECE 4.31%
Accuracy	SVM	4	Accuracy 66.33%

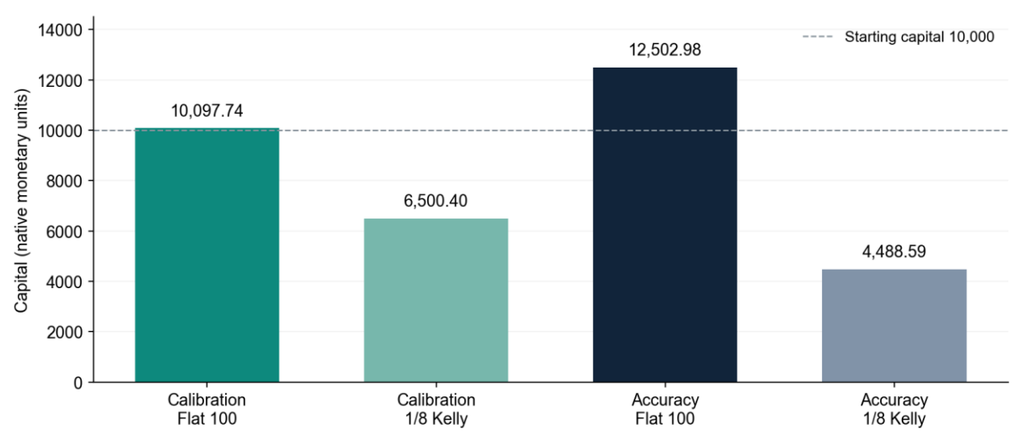
Train: 2014/15–2015/16 → tune: 2016/17 → family selection: 2017/18 → bet: 2018/19.
The two branches use different selected feature sets; the row scores are not joint metrics of one fitted model.

Sources: Data.ipynb | Full paths in notes

NBA: saved 2018/19 betting results

Four rules, each starting with 10,000. Fixed 100 stakes equal 1% of initial capital.

Branch / model	Stake rule	Final capital	Capital return	Trade frequency	Hit ratio
Calibration (LR)	100 fixed	10,097.74	0.98%	85.3%	38.0%
Calibration (LR)	1/8 Kelly	6,500.40	-35.00%	85.3%	38.0%
Accuracy (SVM)	100 fixed	12,502.98	25.03%	88.6%	34.9%
Accuracy (SVM)	1/8 Kelly	4,488.59	-55.11%	88.6%	34.9%



Home value is checked first.
Otherwise check away value.
Kelly uses 1/8 of full Kelly,
limited to available cash.
Both saved Kelly outcomes lose.

Saved notebook results: LR reproduced to cents. SVM replay did not match; its saved results are reported, not independently verified.

Sources: Data.ipynb; nba_evidence.json | Full paths in notes

NBA: nine-season annual robustness

Each season starts a fresh 10,000 account with fixed 100 requests. Annual paths are not one funded continuous portfolio.

CALIBRATION-BRANCH MEAN

-17.34%

Mean annual capital return

ACCURACY-BRANCH MEAN

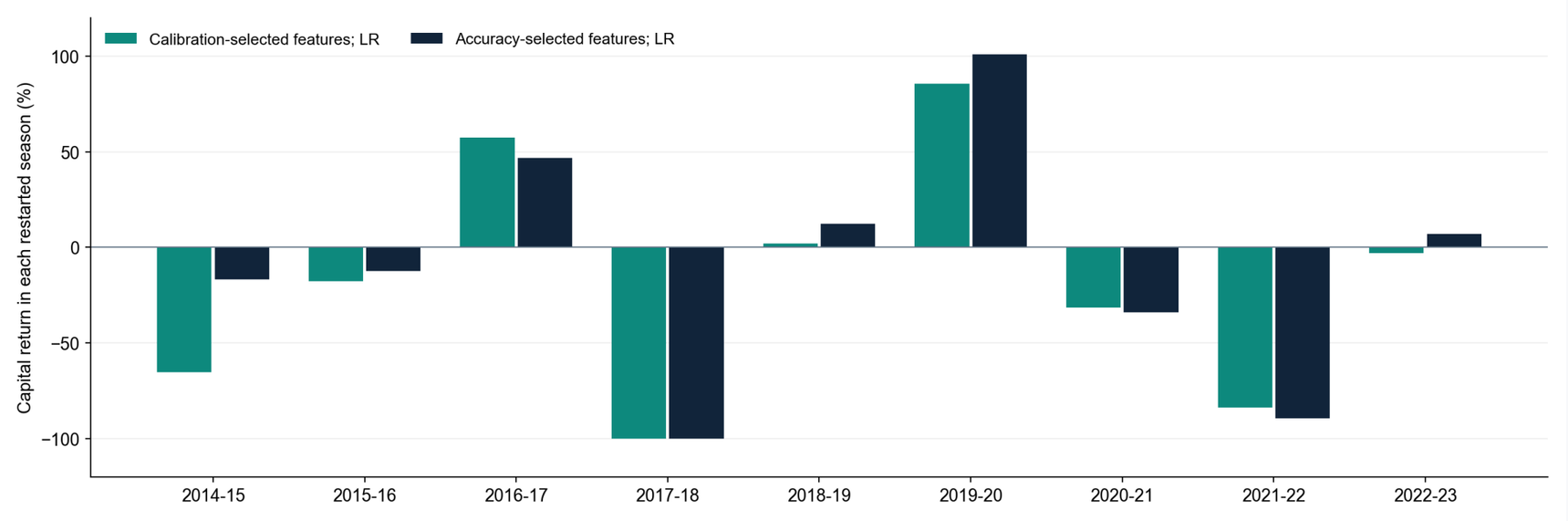
-9.49%

Mean annual capital return

CALIBRATION WINS

3 / 9

Both branches use logistic regression here



This extension reselects features but does not repeat the four-family search. The annual variation does not support a dependable profit advantage.

NBA: what happens if we back the other team?

2017/18: both original fixed-stake accounts ended at zero. Both annual models are logistic regressions.

Original branch	Original → opposite	Hit ratio	Trade frequency	Expected gain / loss	Stake ROI	Max drawdown
Calibration LR	0 → 11,148.07	65.68%	85.12%	27.06 / 46.67	1.25%	29.17%
Accuracy LR	0 → 8,785.30	63.35%	82.23%	27.99 / 47.91	-1.37%	50.76%

Keep the original entry signal; bet on the other team at its own recorded odds.
Start with 10,000; request 100 per bet (1% of initial cash), capped by available cash.
Continue through all 1,075 games, including games after the original account ran out of money.

One full-season inverse profits and one loses. Reversing only the original funded trades loses in both cases: 9,420 and 8,156.

Post-hoc source-method reconstruction. Both conditional 95% ROI intervals include zero; this is not a confirmed edge.
Expected gain/loss use the original model's probabilities; actual quoted opposite returns are not minus the original profit.

Sources: inverse_metrics.csv; audit_summary.json; conditional_uncertainty.csv | Full paths in notes

EPL: the proposal's accuracy–calibration comparison

The first experiment evaluates 374 matches in 2025/26 after separate tuning, calibration, family selection and strategy validation.

Probability model	Accuracy	ECE 10	Log loss
Accuracy-selected SVM	48.13%	6.20%	1.0763
Calibration-selected MLP	44.39%	4.16%	1.0564
Bet365 market benchmark	48.66%	4.36%	1.0191

PRIMARY POLICIES

Hold cash

Neither branch found an admissible positive-return validation threshold.

SECONDARY DIAGNOSTIC: predeclared 5% edge, best archived quotes, 1,000 flat stakes from 100,000

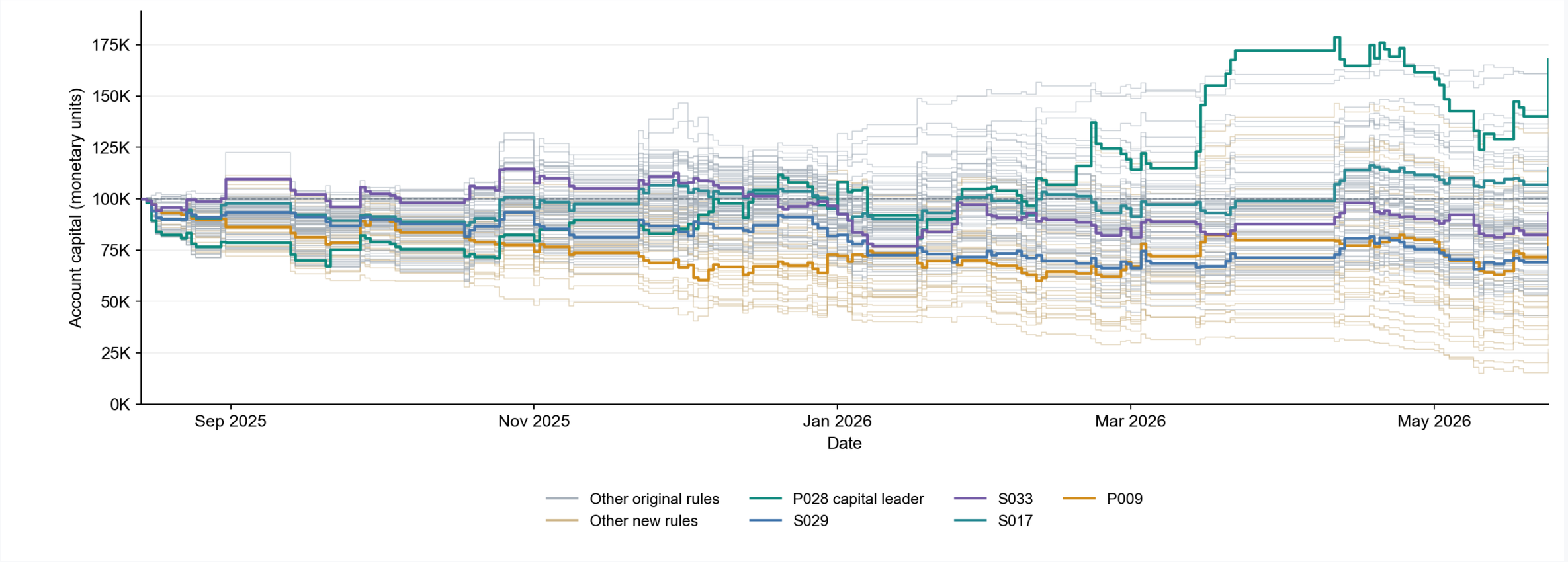
Model	Bets	Profit	Stake ROI	Capital return
SVM	371	-8,330	-2.25%	-8.33%
MLP	333	1,370	0.41%	1.37%

The MLP's +0.41% stake ROI is uncertain: its conditional 95% interval includes zero. The following EPL extensions are exploratory.

Sources: RESULTS.md; classification_scores.csv; metrics.csv | Full paths in notes

EPL: money progression of all 116 strategies

2025/26 is the common season available for every account. Each starts with 100,000 monetary units.



All 84 original and 32 new rules are included. The next model pages label their separate four-year statistics explicitly.

Sources: graph_equity_2025_all.csv | Full paths in notes

EPL: best prediction model: odds + team/player form

Lowest prediction error among the learned models on the same four-year sample. Trading statistics: 2022/23–2025/26, 1,470 matches.

WHAT IT DOES

Starts with bookmaker probabilities, then adjusts them using earlier team results and player form.

ENDING CAPITAL

123,340

Started at 100,000

RETURN ON INITIAL CAPITAL

23.34%

Four-year account

WHEN AND HOW MUCH IT BETS

Choose home, draw or away with the highest estimated return.
Bet only when the model estimates a return above 5%.
Request 1,000 per bet: 1% of the initial 100,000.

Forecast accuracy: 54.15% | Prediction loss: 0.9693

Trading statistic	Result
Bets / trading frequency	1,147 / 78.03%
Hit ratio: bets won	31.04%
Expected gain per trade	852.95
Expected loss per trade	663.70
Profit / ROI on total stakes	23,340 / 2.03%
Largest drop from a prior peak	31.88%

Stakes can be reduced to available opening cash. Expected gain/loss are model estimates, not realized average wins/losses.

Sources: predictions.parquet; metrics_continuous.parquet | Full paths in notes

EPL: simplest model: past result frequencies

No match-specific inputs: every game in a season gets the same three probabilities. Trading statistics: 2022/23–2025/26, 1,470 matches.

WHAT IT DOES

Counts how often home wins, draws and away wins happened in the training data. Uses those fractions for every game.

WHEN AND HOW MUCH IT BETS

Choose home, draw or away with the highest estimated return.
Bet only when the model estimates a return above 5%.
Request 1,000 per bet: 1% of the initial 100,000.

Forecast accuracy: 44.29% | Prediction loss: 1.0700

ENDING CAPITAL

29,760

Started at 100,000

RETURN ON INITIAL CAPITAL

-70.24%

Four-year account

Trading statistic	Result
Bets / trading frequency	1,447 / 98.44%
Hit ratio: bets won	23.01%
Expected gain per trade	1739.52
Expected loss per trade	611.96
Profit / ROI on total stakes	-70,240 / -4.85%
Largest drop from a prior peak	89.49%

Stakes can be reduced to available opening cash. Expected gain/loss are model estimates, not realized average wins/losses.

Sources: predictions.parquet; metrics_continuous.parquet | Full paths in notes

EPL: team model: earlier team performance

Uses team performance and data-coverage controls; no player-performance or odds predictors. Trading statistics: 2022/23–2025/26, 1,470 matches.

WHAT IT DOES

Uses earlier goals, shots on target, corners, points, rest days and previous-season results to estimate home/draw/away chances.

WHEN AND HOW MUCH IT BETS

Choose home, draw or away with the highest estimated return.
Bet only when the model estimates a return above 5%.
Request 1,000 per bet: 1% of the initial 100,000.

Forecast accuracy: 52.93% | Prediction loss: 0.9876

ENDING CAPITAL

119,790

Started at 100,000

RETURN ON INITIAL CAPITAL

19.79%

Four-year account

Trading statistic	Result
Bets / trading frequency	1,241 / 84.42%
Hit ratio: bets won	29.57%
Expected gain per trade	962.54
Expected loss per trade	648.43
Profit / ROI on total stakes	19,790 / 1.59%
Largest drop from a prior peak	40.35%

Stakes can be reduced to available opening cash. Expected gain/loss are model estimates, not realized average wins/losses.

Sources: predictions.parquet; metrics_continuous.parquet | Full paths in notes

EPL: player model: earlier Fantasy Premier League form

Uses previous player records; no team-strength or bookmaker-odds predictors. Trading statistics: 2022/23–2025/26, 1,470 matches.

WHAT IT DOES

Uses earlier FPL points, goals and assists.
Adds minutes and other player statistics.
Looks back over five and ten games.

WHEN AND HOW MUCH IT BETS

Choose home, draw or away with the highest estimated return.
Bet only when the model estimates a return above 5%.
Request 1,000 per bet: 1% of the initial 100,000.

Forecast accuracy: 50.07% | Prediction loss: 1.0218

ENDING CAPITAL

0

Started at 100,000

RETURN ON INITIAL CAPITAL

-100.00%

Four-year account

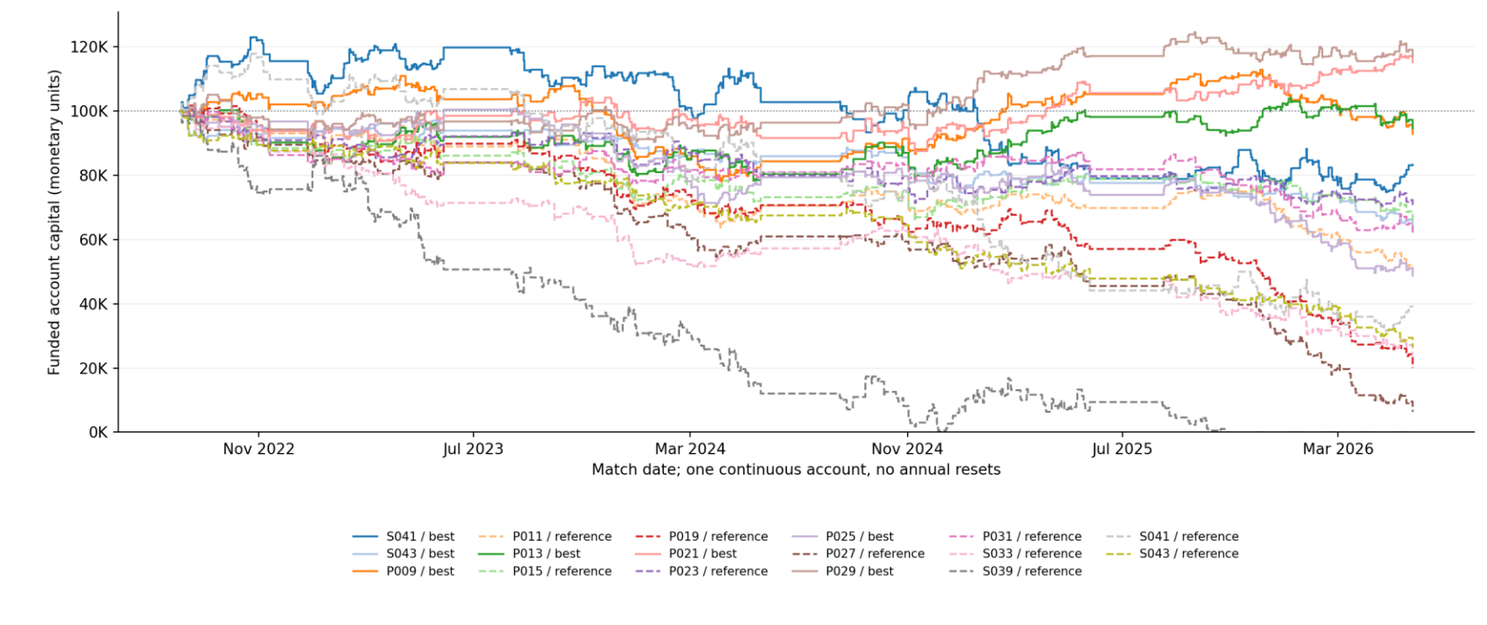
Trading statistic	Result
Bets / trading frequency	944 / 64.22%
Hit ratio: bets won	25.00%
Expected gain per trade	1098.04
Expected loss per trade	618.22
Profit / ROI on total stakes	-100,000 / -10.87%
Largest drop from a prior peak	100.00%

Stakes can be reduced to available opening cash. Expected gain/loss are model estimates, not realized average wins/losses.

Sources: model_metrics_pooled.csv; metrics_continuous.parquet | Full paths in notes

EPL: back both other outcomes after a zero account

17 bankrupt strategy/price contexts, covering 15 strategy IDs. Each inverse starts a new 100,000 account.



FULL-HORIZON PRICED COMPLEMENT

2 / 17 profit at 0% cost

Median ending cash: 62,010
Range: 0–116,968

Profitable after 2% / 5% cost:
2 / 17 and 1 / 17

Controls: 2 / 34 profit.
Profitable cases with 95% ROI intervals including zero: 2 / 2.

Keep the original signal. Split 1,000 total risk (1% of initial cash) across the other two results for equal payouts.
Fund all same-date bundles from opening cash; continue after the original account's bankruptcy.

Hypothetical lays profit in 16/17 cases but assume unobserved lay quotes. This post-hoc study does not verify executable returns.

Sources: metrics.csv; summary.json; inverse_protocol.md | Full paths in notes

What the completed project supports

Calibration, quoted prices and capital risk must be evaluated together. Historical profit alone is not enough.

Question	Verified evidence	Interpretation
Calibration versus accuracy	NBA calibration wins 3/9 annual comparisons. EPL locked primary policies both hold cash.	No stable, established profit advantage.
Predictive benchmark	EPL log loss: raw market 0.9662; best learned model 0.9693 (lower is better).	Prices remain a strong benchmark.
Extra player data	All 6 tested new player-feature variants have worse loss than their matched team controls.	These richer features did not improve forecasts.
Arbitrage and reversals	4/1,470 mathematical arbitrage flags; 0 verified fills. Opposite accounts profit: NBA 1/2; EPL 2/17.	Neither result establishes an executable future edge.

Limitations: exploratory reuse of outcomes, uncertain quote availability and one unreproduced NBA SVM result.

Next test: freeze the model, rules and risk budget; record available quotes before future games; then evaluate both forecasts and funded returns.

Sources: project_review.md; review_checks.json; epl_evidence.json; metrics.csv | Full paths in notes

Appendix: the four NBA betting rules

One bet at most per game; settle sequentially. This is not the EPL joint same-date funding rule.

Forecast branch	Sizing rule	Money requested per trade	Entry rule
Calibration / LR	Fixed	100 per bet = 1% of initial 10,000	Home value first; otherwise away value
Calibration / LR	Eighth Kelly	1/8 Kelly; variable; cap only available cash	Home value first; otherwise away value
Accuracy / SVM	Fixed	100 per bet = 1% of initial 10,000	Home value first; otherwise away value
Accuracy / SVM	Eighth Kelly	1/8 Kelly; variable; cap only available cash	Home value first; otherwise away value

Eighth-Kelly stake = current cash × (1/8) × max(0, (probability × odds − 1) / (odds − 1)).

Do not add an EPL-style 2% cap to these source rules. A positive estimated return is a model claim, not a guaranteed gain.

Sources: Data.ipynb | Full paths in notes

EPL appendix: how much capital each rule uses

Stake means the amount requested for one bet. The next pages state the setting for every strategy.

Stake type	Plain meaning	Example
Fixed 1,000	Same requested amount each bet: 1% of INITIAL 100,000	Cash 80,000 → still request 1,000
1% of current cash	One percent of cash at the start of the match date	Cash 80,000 → request 800
Kelly sizing	Amount varies with estimated advantage and odds; the listed % is a ceiling	A 2% cap means at most 2,000 when cash is 100,000

Kelly example: 100,000 cash, a 40% chance and odds of 3.
Quarter Kelly requests 2,500; a 2% cap reduces the bet to 2,000.

Kelly stake = opening cash × min(cap, Kelly fraction × (probability × odds − 1) / (odds − 1)).

All bets on a date share opening cash. Date caps apply to their combined stakes. Reduce all requests proportionally when needed; never borrow.

Sources: engine.py; all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (1/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
S001–S004	Original SVM forecast	No bets; keep cash	0	No betting
S005 / S006	Original SVM forecast	Estimated return > 5%	S005: 1,000 S006: 1/4 Kelly; ≤2% cash	Best odds Cash available
S007 / S008	Original SVM forecast	Estimated return > 5%	S007: 1,000 S008: 1/4 Kelly; ≤2% cash	Bet365 Cash available
S009–S012	Original neural-network forecast	No bets; keep cash	0	No betting
S013 / S014	Original neural-network forecast	Estimated return > 5%	S013: 1,000 S014: 1/4 Kelly; ≤2% cash	Best odds Cash available
S015 / S016	Original neural-network forecast	Estimated return > 5%	S015: 1,000 S016: 1/4 Kelly; ≤2% cash	Bet365 Cash available
S017 / S018	Team form	Estimated return > 5%	S017: 1,000 S018: 1/4 Kelly; ≤2% cash	Best odds Cash available
S019 / S020	Team form + chance quality	Estimated return > 5%	S019: 1,000 S020: 1/4 Kelly; ≤2% cash	Best odds Cash available
S021 / S022	Team form + player form	Estimated return > 5%	S021: 1,000 S022: 1/4 Kelly; ≤2% cash	Best odds Cash available

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (2/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
S023 / S024	Team + chance quality + player form	Estimated return > 5%	S023: 1,000 S024: 1/4 Kelly; ≤2% cash	Best odds Cash available
S052 / S055	Team + chance quality + player form	Estimated return > 5%	S052: 1% of current cash S055: 1/8 Kelly; ≤1% cash	Best odds Cash available
S056	Team + chance quality + player form	Estimated return > 5%	S056: 1/2 Kelly; ≤2% cash	Best odds Cash available
S025 / S026	Bookmaker odds + team form	Estimated return > 5%	S025: 1,000 S026: 1/4 Kelly; ≤2% cash	Best odds Cash available
S027 / S028	Bookmaker odds + team/chance quality	Estimated return > 5%	S027: 1,000 S028: 1/4 Kelly; ≤2% cash	Best odds Cash available
S029 / S030	Bookmaker odds + team/player form	Estimated return > 5%	S029: 1,000 S030: 1/4 Kelly; ≤2% cash	Best odds Cash available
S031 / S032	Bookmaker odds + team/chance/player form	Estimated return > 5%	S031: 1,000 S032: 1/4 Kelly; ≤2% cash	Best odds Cash available
S053 / S057	Bookmaker odds + team/chance/player form	Estimated return > 5%	S053: 1% of current cash S057: 1/8 Kelly; ≤1% cash	Best odds Cash available

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (3/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
S058	Bookmaker odds + team/chance/player form	Estimated return > 5%	S058: 1/2 Kelly; ≤2% cash	Best odds Cash available
S033 / S034	Past home/draw/away frequencies	Estimated return > 5%	S033: 1,000 S034: 1/4 Kelly; ≤2% cash	Best odds Cash available
S035 / S036	Bookmaker probabilities	Estimated return > 5%	S035: 1,000 S036: 1/4 Kelly; ≤2% cash	Best odds Cash available
S037 / S038	Calibrated bookmaker probabilities	Estimated return > 5%	S037: 1,000 S038: 1/4 Kelly; ≤2% cash	Best odds Cash available
S039	Bookmaker probabilities	Always home win No return filter	S039: 1,000	Best odds Cash available
S040	Bookmaker probabilities	Always draw No return filter	S040: 1,000	Best odds Cash available
S041	Bookmaker probabilities	Always away win No return filter	S041: 1,000	Best odds Cash available
S042	Bookmaker probabilities	Bookmaker's most likely result No return filter	S042: 1,000	Best odds Cash available

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (4/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
S043	Bookmaker probabilities	Bookmaker's least likely result No return filter	S043: 1,000	Best odds Cash available
S044	Team + chance quality + player form	Model's most likely result No return filter	S044: 1,000	Best odds Cash available
S045	Bookmaker odds + team/chance/player form	Model's most likely result No return filter	S045: 1,000	Best odds Cash available
S046 / S047	Average of four team/player forecasts	Estimated return > 5%	S046: 1,000 S047: 1/4 Kelly; ≤2% cash	Best odds Cash available
S048 / S049	Average of eight learned forecasts	Estimated return > 5%	S048: 1,000 S049: 1/4 Kelly; ≤2% cash	Best odds Cash available
S054	Average of eight learned forecasts	Estimated return > 5%	S054: 1% of current cash	Best odds Cash available
S050 / S051	Half combined forecast, half bookmaker	Estimated return > 5%	S050: 1,000 S051: 1/4 Kelly; ≤2% cash	Best odds Cash available
S059	Team + chance quality + player form	Estimated return > 5%	S059: 1/4 Kelly; ≤2% cash	Best odds Date cap 10% cash

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (5/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
S060	Bookmaker odds + team/chance/player form	Estimated return > 5%	S060: 1/4 Kelly; ≤2% cash	Best odds Date cap 10% cash
S061	Average of eight learned forecasts	Estimated return > 5%	S061: 1/4 Kelly; ≤2% cash	Best odds Date cap 10% cash
S062 / S063	Average of eight learned forecasts	Estimated return > 5% Positive return: 6/8 models	S062: 1,000 S063: 1/4 Kelly; ≤2% cash	Best odds Cash available
S064	Average of eight learned forecasts	Estimated return > 5% Only H	S064: 1,000	Best odds Cash available
S065	Average of eight learned forecasts	Estimated return > 5% Only D	S065: 1,000	Best odds Cash available
S066	Average of eight learned forecasts	Estimated return > 5% Only A	S066: 1,000	Best odds Cash available
S067	Team + chance quality + player form	Estimated return > 10%	S067: 1,000	Best odds Cash available
S068	Bookmaker odds + team/chance/player form	Estimated return > 10%	S068: 1,000	Best odds Cash available

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (6/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
S069	Average of eight learned forecasts	Estimated return > 10%	S069: 1,000	Best odds Cash available
S070	Team + chance quality + player form	Estimated return > 5% Odds 1.5–5.0	S070: 1,000	Best odds Cash available
S071	Bookmaker odds + team/chance/player form	Estimated return > 5% Odds 1.5–5.0	S071: 1,000	Best odds Cash available
S072	Average of eight learned forecasts	Estimated return > 5% Odds 1.5–5.0	S072: 1,000	Best odds Cash available
S073	Original SVM forecast	Estimated return > 0%	S073: 1,000	Best odds Cash available
S074	Original SVM forecast	Estimated return > 2.5%	S074: 1,000	Best odds Cash available
S075	Original SVM forecast	Estimated return > 7.5%	S075: 1,000	Best odds Cash available
S076	Original SVM forecast	Estimated return > 10%	S076: 1,000	Best odds Cash available

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (7/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
S077	Original SVM forecast	Estimated return > 15%	S077: 1,000	Best odds Cash available
S078	Original SVM forecast	Estimated return > 20%	S078: 1,000	Best odds Cash available
S079	Original neural-network forecast	Estimated return > 0%	S079: 1,000	Best odds Cash available
S080	Original neural-network forecast	Estimated return > 2.5%	S080: 1,000	Best odds Cash available
S081	Original neural-network forecast	Estimated return > 7.5%	S081: 1,000	Best odds Cash available
S082	Original neural-network forecast	Estimated return > 10%	S082: 1,000	Best odds Cash available
S083	Original neural-network forecast	Estimated return > 15%	S083: 1,000	Best odds Cash available
S084	Original neural-network forecast	Estimated return > 20%	S084: 1,000	Best odds Cash available

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (8/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
P001 / P002	Team form (logistic)	Estimated return > 5%	P001: 1,000 P002: 1/4 Kelly; ≤2% cash	Best odds Cash available
P003 / P004	Team form (logistic)	Estimated return > 5%	P003: 1,000 P004: 1/4 Kelly; ≤2% cash	Bet365 Cash available
P005 / P006	Team form (boosted trees)	Estimated return > 5%	P005: 1,000 P006: 1/4 Kelly; ≤2% cash	Best odds Cash available
P007 / P008	Team form (boosted trees)	Estimated return > 5%	P007: 1,000 P008: 1/4 Kelly; ≤2% cash	Bet365 Cash available
P009 / P010	Player form (logistic)	Estimated return > 5%	P009: 1,000 P010: 1/4 Kelly; ≤2% cash	Best odds Cash available
P011 / P012	Player form (logistic)	Estimated return > 5%	P011: 1,000 P012: 1/4 Kelly; ≤2% cash	Bet365 Cash available
P013 / P014	Player form (boosted trees)	Estimated return > 5%	P013: 1,000 P014: 1/4 Kelly; ≤2% cash	Best odds Cash available
P015 / P016	Player form (boosted trees)	Estimated return > 5%	P015: 1,000 P016: 1/4 Kelly; ≤2% cash	Bet365 Cash available

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: strategy rules (9/9)

Default: choose the highest model-estimated return (probability × odds − 1). Only bet above the listed threshold.

Strategy IDs	Data / model	When it bets	Requested money per bet	Odds / total limit
P017 / P018	Player form + estimated XI/depth (logistic)	Estimated return > 5%	P017: 1,000 P018: 1/4 Kelly; ≤2% cash	Best odds Cash available
P019 / P020	Player form + estimated XI/depth (logistic)	Estimated return > 5%	P019: 1,000 P020: 1/4 Kelly; ≤2% cash	Bet365 Cash available
P021 / P022	Player form + estimated XI/depth (trees)	Estimated return > 5%	P021: 1,000 P022: 1/4 Kelly; ≤2% cash	Best odds Cash available
P023 / P024	Player form + estimated XI/depth (trees)	Estimated return > 5%	P023: 1,000 P024: 1/4 Kelly; ≤2% cash	Bet365 Cash available
P025 / P026	Team + player form/XI/depth (logistic)	Estimated return > 5%	P025: 1,000 P026: 1/4 Kelly; ≤2% cash	Best odds Cash available
P027 / P028	Team + player form/XI/depth (logistic)	Estimated return > 5%	P027: 1,000 P028: 1/4 Kelly; ≤2% cash	Bet365 Cash available
P029 / P030	Team + player form/XI/depth (trees)	Estimated return > 5%	P029: 1,000 P030: 1/4 Kelly; ≤2% cash	Best odds Cash available
P031 / P032	Team + player form/XI/depth (trees)	Estimated return > 5%	P031: 1,000 P032: 1/4 Kelly; ≤2% cash	Bet365 Cash available

Sources: all_strategy_catalog.json | Full paths in notes

EPL appendix: every bankrupt-policy inverse (1/2)

Original ending cash is zero in every row. Main results are funded full-horizon complements with no cost, 2022/23–2025/26.

Strategy / quotes	Ending cash	Stake ROI	Profit hit	Bets / signals; frequency	Expected gain / loss	Max drawdown	Ending: 2% / 5% cost	Synthetic lay ending
S033 / Bet365	24,426	-5.36%	76.01%	1409/1409; 95.85%	142.3 / 387.4	75.63%	19,634 / 12,446	120,882
S039 / Bet365	0	-8.91%	54.44%	1194/1470; 81.22%	368.3 / 416.7	100.00%	0 / 0	176,440
S041 / Best	83,274	-1.14%	68.57%	1470/1470; 100.00%	301.2 / 323.0	39.24%	74,388 / 61,060	140,580
S041 / Bet365	39,139	-4.14%	67.14%	1470/1470; 100.00%	271.5 / 323.1	73.18%	31,133 / 19,124	168,948
S043 / Best	63,774	-2.46%	79.46%	1470/1470; 100.00%	180.8 / 207.3	36.38%	58,478 / 50,535	107,026
S043 / Bet365	26,653	-4.99%	77.96%	1470/1470; 100.00%	155.7 / 207.4	73.41%	22,097 / 15,262	124,838
P009 / Best	92,927	-0.51%	75.43%	1384/1384; 94.15%	199.2 / 354.8	29.63%	86,268 / 76,280	138,377
P011 / Bet365	49,178	-3.94%	74.73%	1290/1290; 87.76%	170.5 / 356.1	50.93%	43,833 / 35,816	145,073
P013 / Best	94,727	-0.39%	76.83%	1351/1351; 91.90%	188.8 / 342.0	21.94%	88,593 / 79,391	138,138

Risk 1,000 TOTAL across two outcomes; start 100,000. Same-date requests share opening cash and can be scaled.
Hit = positive bundle profit; expected gain/loss are model claims. Synthetic lay assumes unobserved lay quotes.

Sources: metrics.csv; identity_diagnostics.csv; inverse_protocol.md | Full paths in notes

EPL appendix: every bankrupt-policy inverse (2/2)

Original ending cash is zero in every row. Main results are funded full-horizon complements with no cost, 2022/23–2025/26.

Strategy / quotes	Ending cash	Stake ROI	Profit hit	Bets / signals; frequency	Expected gain / loss	Max drawdown	Ending: 2% / 5% cost	Synthetic lay ending
P015 / Bet365	66,059	-2.66%	76.41%	1276/1276; 86.80%	163.1 / 344.6	34.18%	60,996 / 53,402	159,942
P019 / Bet365	20,121	-5.87%	73.31%	1360/1360; 92.52%	167.6 / 365.5	80.38%	14,677 / 6,510	118,747
P021 / Best	115,263	1.11%	77.74%	1370/1370; 93.20%	191.2 / 344.9	16.20%	108,878 / 99,300	160,426
P023 / Bet365	70,366	-2.30%	76.38%	1287/1287; 87.55%	164.1 / 346.5	29.82%	65,216 / 57,491	165,887
P025 / Best	48,758	-3.65%	73.75%	1402/1402; 95.37%	194.8 / 359.4	52.02%	42,443 / 32,970	93,065
P027 / Bet365	6,421	-7.05%	72.89%	1328/1328; 90.34%	169.3 / 362.6	93.61%	2,420 / 0	101,274
P029 / Best	116,968	1.26%	76.23%	1346/1346; 91.56%	202.2 / 348.5	13.98%	110,249 / 100,170	162,936
P031 / Bet365	62,010	-3.08%	74.47%	1234/1234; 83.95%	172.5 / 348.9	38.47%	56,767 / 48,903	155,770

Risk 1,000 TOTAL across two outcomes; start 100,000. Same-date requests share opening cash and can be scaled.
Hit = positive bundle profit; expected gain/loss are model claims. Synthetic lay assumes unobserved lay quotes.

Sources: metrics.csv; identity_diagnostics.csv; inverse_protocol.md | Full paths in notes